

Evaluation Study

Ablation study across six retrieval configurations

Model: gemma4:e2b | 14 ground-truth queries | Corpus: 176 dense vectors / 77 BM25 documents

Metric	Dense Only	BM25 Only	Hybrid Static	Hybrid Adaptive	Hybrid + Reranker	Full Pipeline
RETRIEVAL METRICS						
Embedding time ↓	11.65 s	0.00 s	12.17 s	11.83 s	13.30 s	12.15 s
Retrieval latency ↓	0.044 s	0.004 s	0.050 s	0.046 s	0.167 s	0.169 s
Relevance – Text ↑	0.7533	0.6193	0.7502	0.7489	0.7439	0.7462
Relevance – Image ↑	0.4248	0.5858	0.4297	0.4175	0.4283	0.4190
Page Coverage@k ↑	92.86%	89.29%	92.86%	92.86%	89.29%	89.29%
GENERATION QUALITY						
ROUGE-1 ↑	0.5072	0.5119	0.5114	0.5105	0.5005	0.5108
ROUGE-2 ↑	0.1943	0.1913	0.1909	0.1938	0.1847	0.1925
ROUGE-L ↑	0.2897	0.2888	0.2917	0.2904	0.2901	0.2939
BLEU ↑	0.1251	0.1234	0.1245	0.1273	0.1184	0.1242
METEOR ↑	0.3422	0.3601	0.3445	0.3523	0.3295	0.3468
BERTScore F1 ↑	0.8772	0.8760	0.8774	0.8769	0.8752	0.8771
FCD ↓	16.55	17.45	16.16	15.03	19.04	18.29
Faithfulness ↑	89.93%	91.77%	91.20%	90.21%	90.54%	89.86%
GT Coverage ↑	97.14%	93.93%	96.79%	97.38%	98.57%	98.57%
PERFORMANCE						
End-to-end latency ↓	18.41 s	10.38 s	19.89 s	20.50 s	19.12 s	19.28 s
Throughput ↑	0.054 q/s	0.096 q/s	0.050 q/s	0.049 q/s	0.052 q/s	0.052 q/s
RAM usage ↓	0.119 GB	0.269 GB	0.119 GB	0.127 GB	0.118 GB	0.131 GB
GPU utilisation ↓	28.30%	40.85%	28.75%	28.40%	26.43%	26.36%

Configurations

- Dense Only** — semantic vector search only.
- BM25 Only** — lexical BM25 search only.
- Hybrid Static** — BM25 + semantic with fixed 0.5 / 0.5 weights.
- Hybrid Adaptive** — query-aware weight blending.
- Hybrid + Reranker** — static hybrid with cross-encoder reranking.
- Full Pipeline** — adaptive hybrid + reranker + distance filtering.

Metric Notes

- ↓ lower is better | ↑ higher is better.
- FCD** — Faithfulness Cosine Distance; divergence between response and prompt embeddings.
- GT Coverage** — fraction of ground-truth answer tokens present in retrieved context.
- Retrieval metrics are model-independent and computed once per configuration before any LLM call.
- Green cells indicate the best (optimal) value for that metric.